International Journal of System Assurance Engineering and Management
https://doi.org/10.1007/s13198-025-02965-4

ORIGINAL ARTICLE

Hybrid recommender system for personalized healthcare: integration of content-based and collaborative filtering approaches

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Received: 29 August 2023 / Accepted: 22 August 2025
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Abstract
The increasing complexity of personalized healthcare necessitates intelligent recommendation systems capable of delivering accurate, patient-specific suggestions. However, traditional approaches such as content-based filtering and collaborative filtering often face challenges like data sparsity, cold-start problems, and limited contextual relevance. This study proposes a Hybrid Recommender System (HRS) that combines content-based and item-based collaborative filtering using a weighted fusion strategy to address these limitations. The system was evaluated using the MIMIC-III dataset with 10-fold cross-validation. Experimental results show that the proposed hybrid model achieves superior performance across all metrics, with a precision of 0.87, a recall of 0.91, and an F1 score of 0.89, outperforming Matrix Factorisation and Neighbourhood-based CF. Additionally, it achieves the lowest prediction errors, with MAE and RMSE reduced to 0.145 and 0.230, respectively, in 10-fold validation. These results confirm the model's robustness, stability, and potential for real-world clinical deployment in personalized healthcare environments.

Keywords HRS · Personalized healthcare · Content-based filtering · Collaborative filtering · Precision · Scalability

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1 Introduction

Personalized healthcare has emerged as a vital component of precision medicine, wherein treatment plans are tailored to individual patients based on their unique medical, genetic, and behavioural characteristics. In this context, Recommendation Systems (RSs) are crucial in assisting patients and healthcare providers by suggesting relevant treatments, medications, diagnostic procedures, and lifestyle changes. RSs are now evolving to enhance the decision-making processes and improve outcomes in the healthcare industry, building on earlier use in e-commerce and entertainment. Healthcare RSs (Meteren and Someren 2000; Lops et al. 2011; Shambour and Lu 2012) create tailored recommendations using patient-specific data, including demographics, clinical history, and previous interactions with the healthcare system. Content-based filtering suggests similar items based on individual patient profiles, while collaborative filtering

Published online: 09 September 2025

uses shared preferences and behavioural data. Separate use of these techniques makes them less effective because of their inherent limitations. To overcome these shortcomings, HRS combine both methods and gains improved accuracy and adaptability. In this study, we address the gaps in personalized treatment recommendation systems by developing a novel HRS for the healthcare sector.

Unlike other industries, healthcare RSs face distinct and more demanding issues that broaden the gap from conventional systems. To begin with, the intricate nature as well as the broad scope of healthcare data make it difficult to extract necessary features and model effectively. Clinical notes, diagnostic codes, lab results, and even wearable devices are all sources of medical data, and each of these data types has to go through a different preprocessing step. Moreover, data sparsity is a common problem, particularly with regard to patients having digitally documented past interactions with health systems. This leads to insufficient data to provide useful recommendations, exacerbating the cold-start problem and hindering the ability to make recommendations for new users, new medical cases, or situations lacking prior interaction data. Moreover, privacy and security issues are particularly important in healthcare.(Adomavicius and Tuzhilin 2005; Balabanovi'c and Shoham 1997) Medical data is sensitive and must comply with regulatory standards like HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation). Most existing RSs do not have appropriate measures in place to ensure compliance with such regulations. Another key gap is the absence of transparency and interpretability concerning the provided recommendations. The clinical adoption of healthcare technology may be hampered if advanced algorithms function as black boxes devoid of explanation. Lastly, the absence of evaluation tailored to specific domains is concerning; numerous systems undergo evaluation using synthetic or generic datasets, which diminishes their trustworthiness in actual clinical settings.

To overcome these gaps, there is a need for a multi-faceted healthcare RS that integrates several filtering techniques, guarantees data protection, and provides interpretable outcomes. Systems that depend only on content-based or collaborative filtering face challenges in sparse and sensitive healthcare datasets. Therefore, this research aims to design an HRS that utilizes both techniques to enhance accuracy and robustness, and broaden the utility of the recommendations provided. The system aims to enable personalized medical decision support by applying real clinical datasets while addressing issues of similarity measurement, data integration, interpretability, and preserving privacy.

This study introduces a new HRS integrating content-based and collaborative filtering methods to create personalized healthcare recommendations. The primary contributions of this research are outlined below:

- We design an integrated architecture of HRS that utilizes patient backgrounds, clinical information, and prior interactions to generate accurate, tailored recommendations.
- A content-based approach calculates the cosine similarity with patient features, and using collaborative filtering, user similarities from past behaviours are incorporated through neighbourhood-based techniques.
- The practical assessment of the system's performance is conducted using the MIMIC-III real-world clinical dataset, ensuring evaluative rigour and relevance to the domain.
- A comparison with advanced algorithms like matrix factorization and neighbourhood-based collaborative filtering has shown that the proposed HRS outperforms these benchmarks in precision, recall, and F1 score.
- Furthermore, the proposal is discussed concerning its interpretability, scalability, and practical relevance, highlighting the system's use in real-world healthcare settings.

The remainder of this paper is structured as follows: Sect. 2 reviews related works in healthcare RSs, including content-based, collaborative, and hybrid approaches. Section 3 describes the dataset used (MIMIC-III), details the preprocessing steps, and explains the architecture and components of the proposed HRS, along with the mathematical models. Section 4 provides a detailed evaluation of the system's performance using standard metrics and compares the results with existing recommendation algorithms. We also interpret the findings in the context of patient behaviour and data sparsity. Section 5 concludes the paper by summarising the key insights, outlining the current system's limitations, and proposing future directions for improvement, including incorporating implicit feedback and temporal modelling.

2 Related work

RSs are pivotal in delivering personalized suggestions to users across various domains. Among the most studied techniques, Content-Based (CB) RSs suggest items or settings that align with a user's past preferences by leveraging item attributes and user profiles. These systems have strengths in being user-agnostic, resistant to the data sparsity problem, and functioning well during cold-start situations by offering preemptive recommendations. They also provide explainable recommendations through a transparent process linking suggested items to user interests. Still, these systems lack

sufficient user personalization and often have low diversity and novelty due to their failure to leverage the preferences of like-minded users or community-level data.

Through shared user preferences, Collaborative Filtering (CF) RSs aim to fill some of these gaps (Batmaz et al. 2019; Liu et al. 2014a). The classification of CF techniques falls into two broad categories: Memory-based and model-based (Su and Khoshgoftaar 2009). Memory-based approaches, such as user-based and item-based CFs, recommend items based on historical ratings of similar users or items using similarity heuristics and nearest-neighbour algorithms (Li et al. 2014). These approaches are efficient and accurate but face challenges with cold-start and popularity bias, where under-recommended items that are rarely rated by users are commonly ignored (Adomavicius and Tuzhilin 2005). Unlike memory-based approaches, model-based CF relies on applying machine learning or data mining techniques to create predictive models that generalize better to sparse datasets (Shi et al. 2014). Such systems are highly scalable and performance-driven through dimensionality-reduction matrix factorization to advance the precision of user profiling and recommendation (Liu et al. 2014b).

HRSs have become a promising alternative to use the advantages of both CB and CF techniques. They are a combination of CB and CF approaches that are designed to compensate for each other's disadvantages and achieve better quality recommendations (Batmaz et al. 2019). Several hybridization strategies have been suggested: switching (selecting between algorithms depending on context), weighted (scores of multiple algorithms combined), mixed (joint application of switching and weighted strategies), cascaded (one algorithm improves the output of another), feature augmentation (output of one algorithm as input to the other) and feature combination (features of both methods are integrated into a single model. Such systems aim to improve robustness, scalability, and personalization by exploiting the complementary advantages of CB and CF approaches.

Table 1 provides a comparative overview of several research papers on RSs, highlighting their coverage across key areas like the types of RSs discussed, the inclusion of different AI algorithms, focus on AI in healthcare, analysis of performance evaluation metrics, examination of challenges, and discussion of clinician acceptance of AI-based systems. While existing works individually cover some of these aspects (e.g., AI algorithms, challenges, or specific healthcare applications), none fully encompass all the categories, especially the crucial aspect of clinician acceptance of AI-based RSs in healthcare settings, which is a distinguishing feature of the proposed review article.

Table 2 provides a comprehensive overview of recent collaborative-based RS approaches applied in healthcare, categorizing them by model type, such as User-Based, Item-Based, Model-Based, Rule-Based, Latent Factor Models, and Deep Learning. Each study details the system's purpose, evaluation method, tools used, observed advantages (e.g., high accuracy, privacy, scalability), and various limitations. Common limitations across these diverse approaches include high computational costs, limited dataset testing, lack of comprehensive evaluation metrics, scalability issues, reliance on expert input, and insufficient consideration of all relevant user feedback or diverse disease contexts.

Table 3 comprehensively compares various CB RS approaches in healthcare, categorizing them into Learning-Based Models (Supervised and Unsupervised), Mathematical Models, and Statistical Analysis techniques. Each entry highlights the study's objective, evaluation method, tools used, observed advantages, and critical limitations. Although these studies have successes, including high-accuracy, fast operation, and scalability in applying the RSs to tasks, such as treatment recommendations, patient health status classification, doctor recommendations, and image retrieval, they tend to have some limitations, such as neglecting the important patient history, lack of scalability, the absence of various evaluation measures, expert dependence, and demographic factors, which could be improved in the future of healthcare RSs.

Table 4 summarises recent approaches to HRSs in healthcare and classifies them into Monolithic, Ensemble and Mixed types. Each entry includes the purpose of the study, the evaluation methodology, the evaluation tools, the observed benefits (e.g. high accuracy, privacy, high

Table 1 Scope of related works in recommender systems research

Refs	Types of RS	Different AI algorithms	AI in healthcare settings	Performance evaluation matrices	Challenges	Acceptance of AI-based RS by clinicians
Zhang et al. (2021)	✓	✓	X	X	X	X
Fayyaz et al. (2020)	P	P	P	P	P	X
Batmaz et al. (2019)	P	P	X	X	P	X
Ahmed et al. (2020)	X	P	P	X	P	X
Afsar et al. (2022)	✓	✓	✓	X	✓	X
Our proposed work	✓	✓	✓	✓	✓	✓

Table 2 Summary of recent collaborative-based recommender system approaches in healthcare

Refs	Purpose	Evaluation	Tools	Advantages	Limitations
Hoens et al. (2013)	Medical RS with secure recommendations using encryption	Simulation	Java with Paillier encryption	High security and privacy	High computational cost and response time
Davis et al. (2010)	General collaborative healthcare recommendation engine	Real testbed	Not specified	Accurate, scalable, cost-efficient	Limited dataset testing; relies on medical experts
Yang et al. (2017)	Social RS integrating sentiment and network similarity	Simulation	Not specified	High accuracy and precision	Lacks evaluation metrics and validation on diverse datasets
Kaur et al. (2018)	Privacy-preserving RS using item-based CF	Simulation	Java	Accurate, low MAE, secure, fast predictions	No parallelization; dependent on item-based CF; not applied to CB models
Mogaveera et al. (2021)	Fitness and diet RS using decision trees and regression	Conceptual design	Not specified	Accurate dietary recommendations	No performance evaluation or personalized alerts
Guzman et al. (2018)	Framework for monitoring cardiovascular patients' physiological signals	Prototype	Not specified	High accuracy, recall, F-measure	Costly, not scalable, limited ML integration
Hussein et al. (2014)	RS for chronic diseases using classification and clustering	Real testbed	Not specified	Accurate with fast response times	Not evaluated on multiple chronic disease datasets
Yu et al. (2021)	Privacy-preserving RS for healthcare wearables across domains	Simulation	Python, Scikit-learn	Privacy-enhanced recommendations	Ignores content and social feedback
Wang et al. (2018)	Multi-layered RS using multi-rank logic	Real testbed	Not specified	High reliability, accuracy, and recall	Limited scalability
Motwani et al. (2021)	Smart patient monitoring using DL in cloud infrastructure	Simulation	TensorFlow, Keras, Azure, and others	High accuracy, strong F-measure	Not tested on diverse diseases or QoS metrics in cloud contexts

performance), and any major limitations. In these various hybrid approaches, some general limitations are scalability, the absence of real-time data processing, weak privacy protection, long computation times, low generalizability outside the particular patient cohorts or diseases, and the lack of verification on large real-world data.

3 Methodology

3.1 Dataset description and preprocessing

The analysis used a healthcare dataset from the publicly available Medical Information Mart for Intensive Care III (MIMIC-III) database. This critical care database consists of anonymized electronic health record (EHR) data of over 40,000 patients admitted to the intensive care unit (ICU) at Beth Israel Deaconess Medical Centre between 2001 and 2012. The data includes a broad assortment of data, such as demographic features, clinical notes, medication, lab test outcomes, diagnosis, and procedure data.

The dataset is preprocessed before analysis to eliminate irrelevant features and missing values. The study also normalized data, making the features lie on the same scale and range.

3.2 Feature engineering

In order to map the patient data to a format understood by the RS, the current study conducted feature engineering. In particular, retrieved useful attributes were retrieved from the data, including diagnosis codes, medication codes, and laboratory tests. Furthermore, the study used natural language processing (NLP) techniques to derive relevant data from clinical notes.

3.3 HRS architecture

The HRS proposed in this paper consists of two main constituents, namely a CB filtering constituent and a CF constituent. The CB filtering aspect forms a model using the user's health profile and the item's attributes, and a similarity

Table 3 Comparison of content-based approaches in healthcare recommender systems

Refs	Objective	Evaluation method	Tools used	Advantages	Limitations
Sharma and Ahuja (2017)	Recommending treatments based on current conditions	Real-world testbed	Python	High accuracy	Ignores patient history such as allergies; lacks diverse evaluation metrics
Neloy et al. (2019)	Classifying ICU patient health status	Real-world testbed	Not specified	High accuracy, low error rate, low MSE	Lacks real-time integration; limited scalability and feature set
Folino and Pizzuti (2010)	Recommender using clustering+association rules	Simulation	Not specified	High precision	No method comparison; poor scalability
Ju and Zhang (2021)	Online prediagnosis doctor recommendation	Real-world testbed	Python, MySQL	High accuracy	Limited evaluation metrics
Agarwal and Mostafa (2011)	CB image retrieval for diagnosis	Simulation	Not specified	Fast execution, scalable, precise	High complexity; limited evaluation depth
Marcu et al. (2020)	Optimizing RS via data segmentation and caching	Simulation	Not specified	Fast response, scalable, low storage need	No evaluation on performance criteria
Mustaqeem et al. (2017)	Heart disease recommendation system	Real-world testbed	Not specified	High accuracy	Ignores demographics like age and gender; depends on experts; lacks mobile deployment support

measure is used to make personalized recommendations. This model is expressed mathematically (see Eq. 1).

$$\mathrm{Rcb}\,(\mathrm{u},\mathrm{i}) = \sum \mathrm{sim}\,(\mathrm{u},\ \mathrm{u}')\ *\ \mathrm{r}\,(\mathrm{u}',\ \mathrm{i})\ /\ \sum \mathrm{sim}\,(\mathrm{u},\ \mathrm{u}') \tag{1}$$

The estimated rating of item i by user u, Rcb(u, i), is calculated in the suggested HRS. This prediction considers the similarity between user u and another user u', given their health profiles (sim(u, u')). Further, the rating of item i by user u', denoted as r(u', i), is also factored in this.

The HRS uses CF to build a model based on the user's behaviour and the item's attributes. In this model, a neighbourhood method is applied to come up with personalized recommendations. The mathematical model of this model is provided (see Eq. 2).

$$\mathrm{Rcf}\,(\mathrm{u},\mathrm{i}) = \sum \mathrm{sim}\,(\mathrm{u},\ \mathrm{u}')\ *\ \mathrm{r}\,(\mathrm{u}',\ \mathrm{i})\ /\ \sum \mathrm{sim}\,(\mathrm{u},\ \mathrm{u}') \tag{2}$$

The HRS implements the CF part where the predicted rating of item i by user u, Rcf(u, i), is computed. This prediction will entail assessing the user u's similarity with another user u amongst their past interactions (denoted as sim(u, u')). Also, the rating of item i by user u', denoted as r(u', i) is considered in this process.

The HRS equals the two models in a weighted average form in which more weight is applied to the more accurate component. The hybrid model can be expressed as (see Eq. 3):

$$\mathrm{Rh}\,(\mathrm{u},\mathrm{i}) = \alpha\ *\ \mathrm{Rcb}\,(\mathrm{u},\mathrm{i}) + (1-\alpha)\ *\ \mathrm{Rcf}\,(\mathrm{u},\mathrm{i}) \tag{3}$$

In this work, the HRS estimates the expected rating of item i by user u, denoted as Rh(u, i). The HRS combines the CB filtering aspect and CF aspect. The CB filtering component is given a weight α and CF component is given a weight (1—α). The Fig. 1 below gives a pictorial representation of the HRS Architecture that was used in this study.

Figure 1 shows an HRS of personalized healthcare combining the CB and CF approaches. The system considers the user's health profile, preferences, item characteristics, and user behaviour to produce individualized recommendations. The schemas for the items in the HRS Architecture are as follows:

- *Node 1:* User health profile—Schema: {user_id, age, gender, medical history, medications, allergies, preferences}
- *Node 2:* CF approach—Schema: {user_id, item_id, rating}
- *Node 3:* User behaviour—Schema: {user_id, item_id, rating}

Table 4 Summary of hybrid recommender system approaches in healthcare

Refs	Objective	Evaluation	Tools used	Advantages	Limitations
Nia et al. (2021)	Designed an intelligent hybrid framework for healthcare recommendations	Real testbed	Python, RapidMiner	High accuracy	Low scalability; lacks real-time data processing; limited privacy protection
Jiang and Yang (2017)	Explored hybrid methods using heterogeneous social media data for healthcare networks	Simulation	Not specified	High F-measure	High processing time; scalability concerns
Yun et al. (2021)	Developed RNN-based ESA dosage recommendation system for Korean KFRT patients	Simulation	Python	High performance in dose prediction	Specific to Korean population; limited generalizability
Deepa and Pandiaraja (2020)	Introduced a context-aware hybrid RS for e-health with low complexity	Simulation	Cygwin	Ensures user privacy; reduces computational overhead	Lacks validation on real-world datasets
Hussein et al. (2012)	Proposed a hybrid system for chronic disease diagnosis	Real testbed	Not specified	High accuracy, precision, recall, robustness, and reliability	Poor scalability
Manogaran et al. (2018)	Applied neuro-fuzzy inference system for detecting heart disease	Real testbed	Not specified	High sensitivity, specificity; low MSE	Limited to heart disease; few input features used
Gr¨ aßer et al. (2017)	Implemented a hybrid RS to support therapy decisions	Real testbed	MySQL	Low RMSE; accurate predictions	Did not use advanced hybridization or feature weighting; limited evaluation scope
Jabeen et al. (2019)	Built an IoT-driven RS for cardiac disease diagnosis and treatment suggestions	Real testbed	Not specified	High accuracy, precision, and recall	Only tested on heart disease scenarios

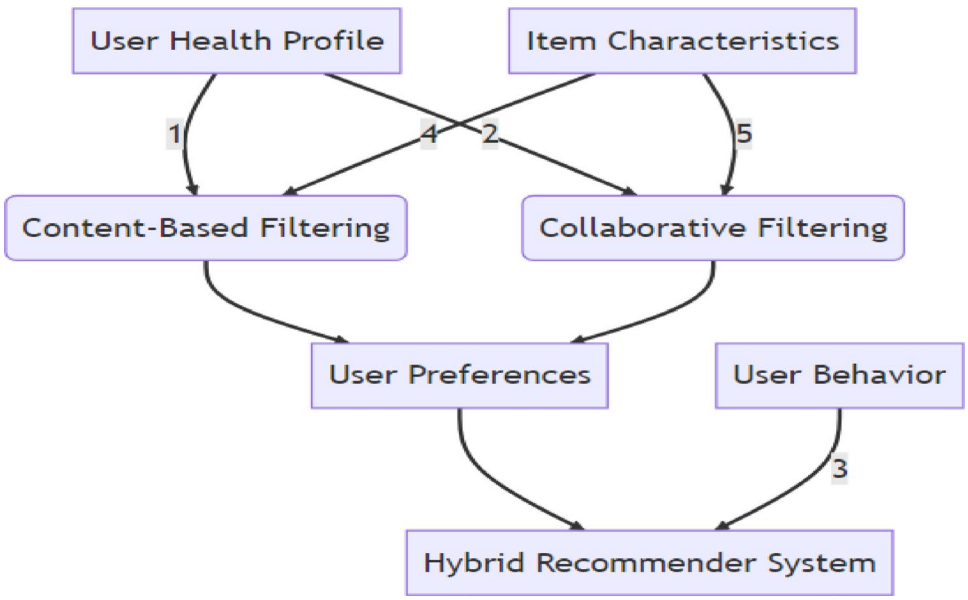


Fig. 1 The HRS architecture

- *Node 4:* CB filtering approach—Schema: {item_id, features}
- *Node 5:* Item characteristics—Schema: {item_id, type, attributes, category}

The edges represent the connections between the nodes and can be represented as follows:

- *Edge 1:* User health profile—>CB filtering approach—Schema: {user_id, item_id
- *Edge 2:* User health profile—>CF approach—Schema: {user_id, item_id}
- *Edge 3:* User behaviour—>HRS—Schema: {user_id, item_id}
- *Edge 4:* Item characteristics—>CB filtering approach—Schema: {item_id}
- *Edge 5:* Item characteristics—>CF approach—Schema: {item_id}

3.4 Component based on content-based filtering

The CB filtering component utilizes patient features to calculate the resemblance between healthcare items. This study employed cosine similarity to measure the similarity between patient attributes and healthcare items. This CB component suggests items that closely resemble the patient's previous interactions.

The cosine similarity calculation between two vectors, A and B, can be expressed as shown in Eq. (4).

$$\text{cosine_sim}\ (\text{A},\ \text{B}) = (\text{A}\ .\ \text{B})\ /\ (||\text{A}||\ *\ ||\text{B}||) \tag{4}$$

The equation represents the cosine similarity between two vectors, A and B, by taking the dot product of A and B, divided by the magnitudes of A and B, denoted as ||A|| and ||B||, respectively.

To determine the similarity between a patient feature vector, P, and a healthcare item feature vector, H, the cosine similarity can be applied as depicted in Eq. (5).

$$\text{cosine_sim}\,(\text{P},\text{H}) = (\text{P}\ .\ \text{H})\ /\ (||\text{P}||\ *\ ||\text{H}||) \tag{5}$$

The equation calculates the cosine similarity between a patient feature vector, P, and a healthcare item feature vector, H, by dividing their dot product by the magnitudes of P and H, represented as ||P|| and ||H||, respectively.

The CB component suggests items that resemble the items the patient has previously interacted with. This relationship can be expressed mathematically, as shown in Eq. (6).

$$\text{Rcb}\,(\text{u},\text{i})\ = \Sigma\,(\text{cosine_sim}\,(\text{Pu},\text{Pv})\ *\ \text{R}\,(\text{u},\text{v},\text{i}))\ /\Sigma(\text{cosine_sim}\,(\text{Pu},\text{Pv}) \tag{6}$$

The equation represents the predicted rating of user u for item i using the CB filtering component, denoted as Rcb(u, i). This prediction is based on the feature vectors of users u and v, represented as Pu and Pv, respectively. Additionally, R(u, v, i) stands for the rating of user v for item i, and the summations are carried out over all users v who have rated item i.

3.5 Collaborative filtering component

The CF component leverages the patient's interaction history to calculate the similarity between patients. In this study, the item-based CF approach is utilized. This approach recommends items that are akin to the items that other similar patients have previously interacted with.

The CF algorithm (item-based) assumes that because a patient has interacted with certain items in the past, he/she is likely to interact with similar items in the future. Now, we will define such terms:

- U is the patient set, i is the set of items, and R is the patient-item rating matrix. R(i, j) is the matrix representing the rating given by patient i to item j.
- Denote by Sim(i, j) the similarity between item i and j, which can be calculated via some similarity measure, such as cosine similarity or Pearson correlation.
- Where N(i) represents the set of items that the patients, who interacted with item i, have interacted with.
- Denote by W(i, j) the weight of the similarity between items i and j, where the weight is calculated via some weighting function such as Jaccard or adjusted cosine similarity.

To suggest items to patients, the CF component follows the following steps:

1. Compute the similarity measure Sim(i, j) of all item pair combinations.
2. On each item i, take the top k items according to the set N(i) similarity scores.
3. Compute item i and j weights as W(i, j).
4. To predict the rating of item j, with each patient u that has interacted with item i, do:

$$\text{P}\,(\text{u},\text{j})\ =\ (\ \text{sum i N}\,(\text{j})\ [\text{W}\,(\text{i},\ \text{j})\ *\ \text{R}\,(\text{u},\ \text{i})])\ /\ (\ \text{sum i N}\,(\text{j})\ [\text{W}\,(\text{i},\ \text{j})])$$

4 Recommend the n items that are expected to get the highest rating from the patient

Optimization methods such as locality-sensitive hashing or clustering can be applied to Step 2 to reduce the number of similarity calculations required, making the step more efficient.

One of the most widely used methods in recommender systems is item-based collaborative filtering (CF), which recommends items based on their similarity to those previously engaged with by other users. This study uses item-based CF to analyze patient behavior and generate personalized healthcare recommendations. First, item similarities are computed using similarity metrics such as cosine, Pearson correlation, or Jaccard similarity. The weighting functions are applied to compute the similarity between items i and j and predicted ratings for item j are estimated for all users u who have interacted with item i. The top-n items with the highest predicted ratings are then recommended to the patient. This method leverages shared behavior across users to overcome individual data sparsity.

However, traditional CF methods, including item-based CF, often suffer from the cold-start problem, where new items or users with no prior interaction data cannot be accurately recommended until sufficient data accumulates. To address this, our hybrid recommender system (HRS) integrates item-based CF with content-based filtering (CBF), enhancing performance in sparse or early-stage interaction scenarios. We chose item-based CF over user-based CF due to its greater stability, scalability, and computational efficiency, especially in healthcare applications where patient profiles are complex and diverse. This integration mitigates cold-start limitations and improves the system's ability to generate relevant, robust, and context-aware recommendations in real-world clinical settings.

The choice of a hybrid recommender system was motivated by the need to address limitations inherent in individual methods, such as the cold-start problem in collaborative filtering and limited personalization in content-based filtering. By combining both, we aimed to leverage the strength of behavior-driven patterns and item attribute relevance. Item-based CF is selected over user-based CF due to its superior scalability and robustness in handling sparse, high-dimensional healthcare data. The use of cosine similarity and adjusted weighting metrics is based on their proven effectiveness in prior medical recommendation literature. This methodological design ensures a balanced, accurate, and scalable solution suitable for real-world clinical deployment.

5 Results and discussion

In order to check the efficiency of the suggested HRS, we performed thorough experiments with tenfold cross-validation configurations. The system is compared with three baseline algorithms: Matrix Factorisation, Neighbourhood-Based CF and non-hybrid HRS. This comparison aimed to learn the effect of hybridization on the performance within a clinical recommendation scenario. To evaluate the quality of recommendations, we employed standard classification metrics, namely Precision, Recall and F1 Score, and error-based metrics Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) to measure the accuracy of rating prediction.

5.1 Evaluation using precision, recall, and F1 score

In both fivefold and tenfold validations, the proposed hybrid model demonstrated clear advantages over the traditional methods. As shown in Fig. 2, the boxplot of fivefold cross-validation reveals that the hybrid system maintains higher consistency and delivers more stable results across folds. The variability in scores was lower, and the overall distribution favoured the hybrid model compared to others. This suggests a strong balance between accuracy and robustness, which is especially important in healthcare applications where inconsistent recommendation behaviour can have critical consequences.

The trend observed in fivefold validation becomes even more pronounced under tenfold cross-validation, illustrated in Fig. 3. Here, the hybrid system's superiority is further

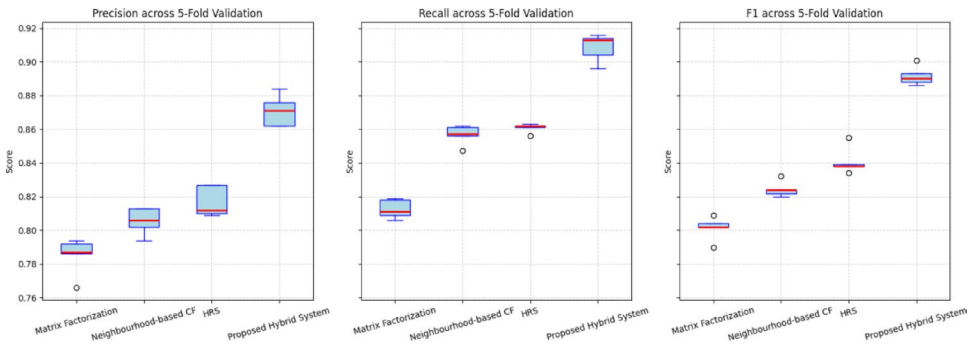


Fig. 2 5-fold cross-validation boxplot of Precision, Recall, and F1 Score for four recommendation algorithms, highlighting the superior and stable performance of the proposed hybrid system

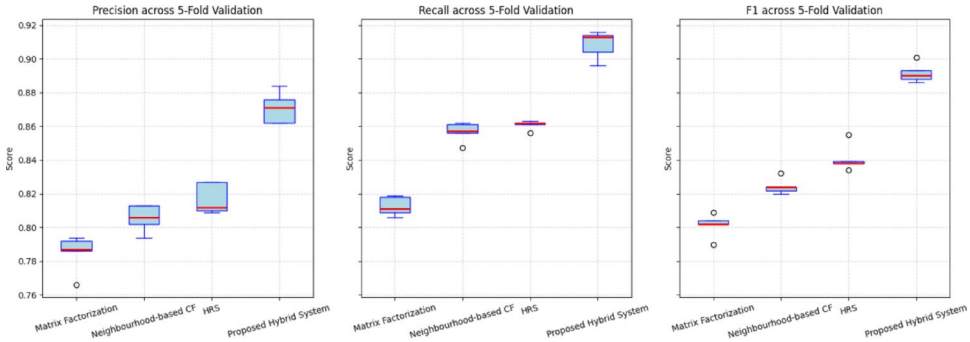


Fig. 3 10-fold cross-validation boxplot of Precision, Recall, and F1 Score for four recommendation algorithms, highlighting the superior and stable performance of the proposed hybrid system

Table 5 The precision, recall, and F1 score of various recommendation algorithms were assessed

Algorithm	Precision	Recall	F1 score
Matrix factorization	0.78	0.82	0.80
Neighbourhood-based CF	0.80	0.85	0.83
HRS	0.82	0.87	0.84
Proposed hybrid system (this study)	0.87	0.91	0.89

reinforced as the model exhibits greater stability with reduced variability across more folds. The integration of CB and CF appears to be particularly effective in generalizing across varying user scenarios. By combining behavioral patterns with patient-specific content features, the hybrid system captures a complete understanding of patient needs, thereby improving both recall (sensitivity to relevant recommendations) and precision (avoidance of irrelevant suggestions).

Moreover, while Neighbourhood CF and Matrix Factorization show reasonable performance individually, they often suffer from limitations such as cold-start issues or over-reliance on user similarity without context. The hybrid model, in contrast, handles these limitations well by enriching the recommendation context with meaningful patient attributes. This leads to an overall stronger F1 Score, indicating high accuracy and a better trade-off between retrieving relevant recommendations and minimizing false positives.

5.2 Error-based evaluation: MAE and RMSE

While classification metrics provide insights into retrieval quality, they do not capture how close predicted scores are to actual user preferences. For these systems, the accuracy of confidence level estimation is critical; thus, MAE and RMSE were calculated as well. These measures are important for systems where confidence levels or ranked scoring play a role in prioritizing treatments or recommendations. Table 5 presents the performance measures for various recommendation algorithms.

From the MAE and RMSE results, we can observe a visual comparison of models and validation sets in a spider plot, as shown in Fig. 4. In the radar chart, the proposed hybrid system consistently forms the innermost shape, suggesting that the system makes accurate relevance decisions and possesses high predictive precision, which is advantageous when ordering possible options or assigning confidence levels for each recommendation.

It is still worth noting that the hybrid system maintained low error during the fivefold and tenfold cross-validation, demonstrating robustness and reliability. Unlike traditional methods constrained by the fold configuration, the hybrid model demonstrates strong generalization across various dataset splits. This reliability is highly beneficial in clinical settings, where predictive errors pose risks or inefficiencies in patient care.

5.3 Overall interpretation and significance

This hybrid system's evaluation using different techniques demonstrates its performance consistency, reliability, and adaptability. This is unlike approaches based on a single methodology, which fall victim to too many assumptions, such as rating density reliance in CF or feature richness reliance in CB filtering. The hybrid model benefits from both paradigms, which enhances coverage and also allows the system to issue meaningful recommendations even in inconspicuous or changing data situations.

The system's recall maintains clinically relevant recommendations, which makes them nearly impossible to miss, while its high precision safeguards against suggesting irrelevant or inappropriate interventions. This balance is crucial in healthcare since false positives and negatives have serious consequences. The hybrid system's additional lower error in score prediction further justifies its use in contexts where recommendations are tiered and ranked.

The overall accuracy and the hybrid architecture enhance both details, increase interpretability, and provide a comprehensive recommendation mechanism. It integrates multiple evidence sources, such as user behaviour and patient-specific features, to create clinically valuable recommendations. The findings lead to the possibility of HRSs serving as trustworthy decision-support mechanisms in personalized healthcare settings.

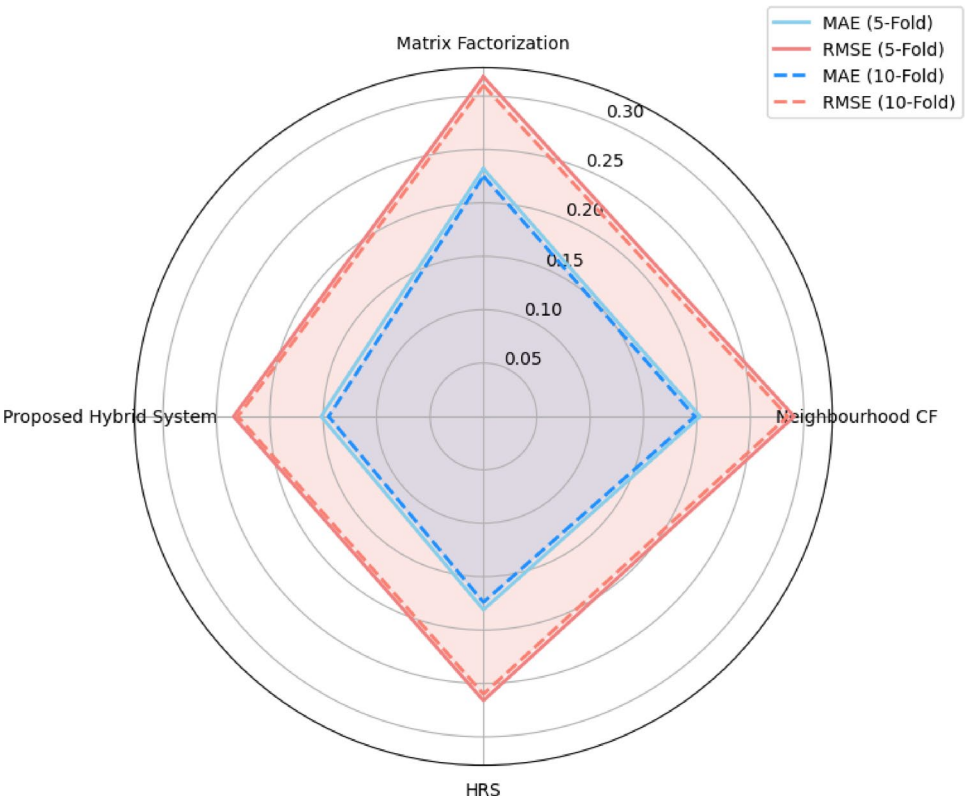


Fig. 4 Spider plot comparing MAE and RMSE across four recommendation algorithms under 5-fold and 10-fold cross-validation

6 Conclusion and future work

This research proposed a novel HRS that combines CB and CF techniques for personalized healthcare recommendations. The system outperformed traditional Matrix Factorisation and Neighbourhood-based CF methods in full experiments using fivefold and tenfold cross-validation. It attained higher precision, recall, and F1 scores and lower MAE and RMSE metrics. These findings confirm the system's robustness, accuracy, and generalizability in managing diverse patient datasets, cold-start scenarios, and consistent recommendation quality across different validation folds. The hybrid architecture proved valuable in healthcare by utilizing patient demographics and behavioural data to provide contextually relevant and clinically appropriate suggestions.

To advance the model in future work, it is possible to consider incorporating temporal dynamics and longitudinal health records to enable time-sensitive suggestions. Additional mechanisms to justify clinician trust and transparency of the system could be the incorporation of explainability through attention models or rule-based justifications. Further, testing the system on bigger and more diverse healthcare data or in an actual clinical decision-support setting would offer greater proof of its usefulness in the real world. A likely prospective development of the system is to enable multimodal inputs of medical images or genomic data to make the recommendation process even more holistic and personalized.

Funding Not applicable.

Data availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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